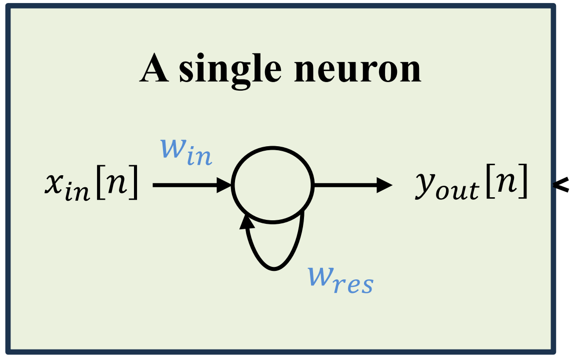
The Linearized RC/ESN

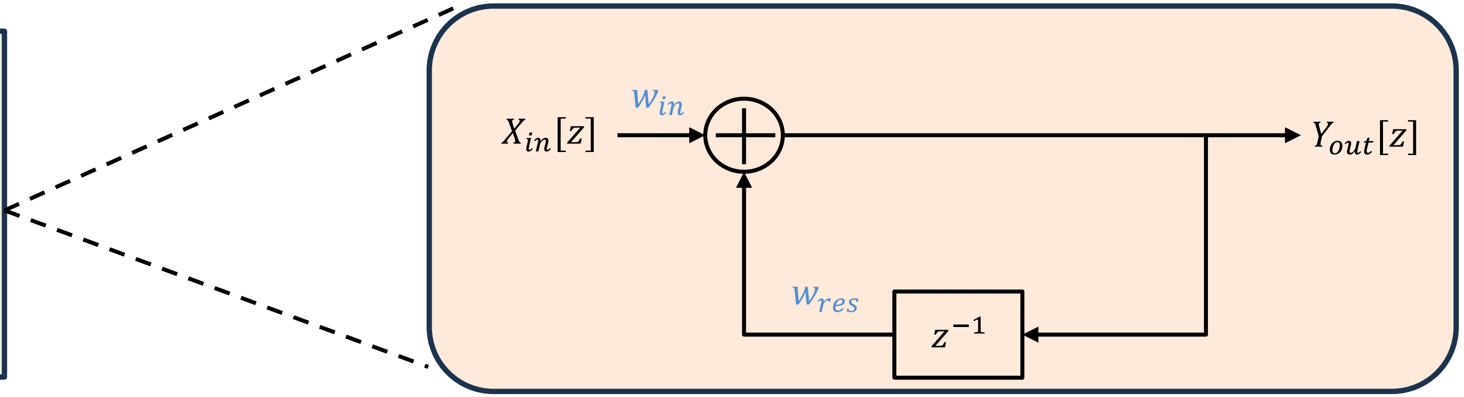
Consider the "linear" ESN for tractable analysis:

$$\boldsymbol{s}_{res}[n] = \mathbf{W}_{res}\boldsymbol{s}_{res}[n-1] + \mathbf{W}_{in}\boldsymbol{x}_{in}[n]$$

$$\boldsymbol{y}_{out}[n] = \mathbf{W}_{out}\boldsymbol{s}_{res}[n]$$

A single neuron in the reservoir with a self loop implements the basic first-order IIR filter





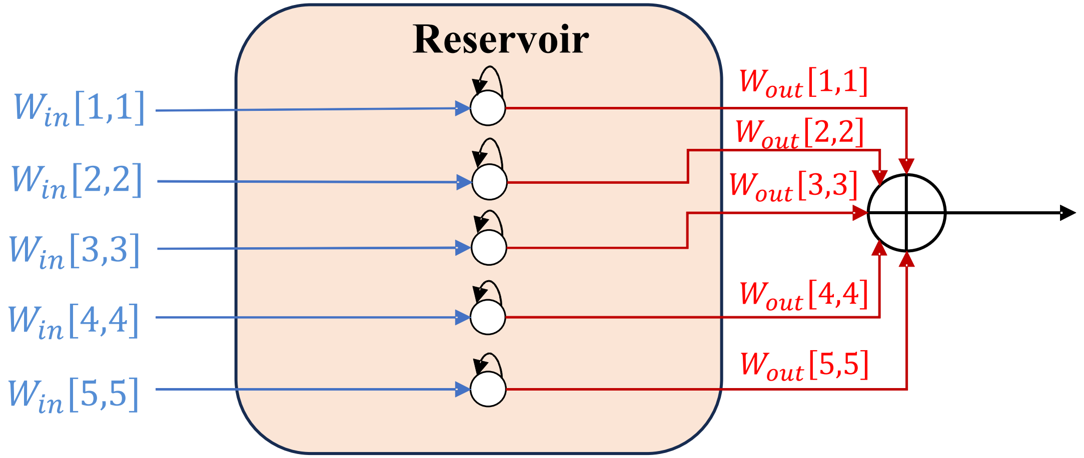
$$y_{out}[n] = w_{res}y_{out}[n-1] + w_{in}x_{in}[n]$$

$$H(z) = \frac{Y_{out}(z)}{X_{in}(z)} = \frac{w_{in}}{1 - w_{res}z^{-1}}$$

S. Jere, R. Safavinejad, L. Zheng and L. Liu, "Channel Equalization Through Reservoir Computing: A Theoretical Perspective," in *IEEE Wireless Communications Letters*, vol. 12, no. 5, pp. 774-778, May 2023.

The Non-Interconnected Linearized ESN

Consider K non-interconnected linearized neurons in the reservoir



Transfer function of k^{th} neuron:

$$H_k(z) = \frac{W_{in}[k,k]}{1 - W_{res}[k,k]z^{-1}}$$

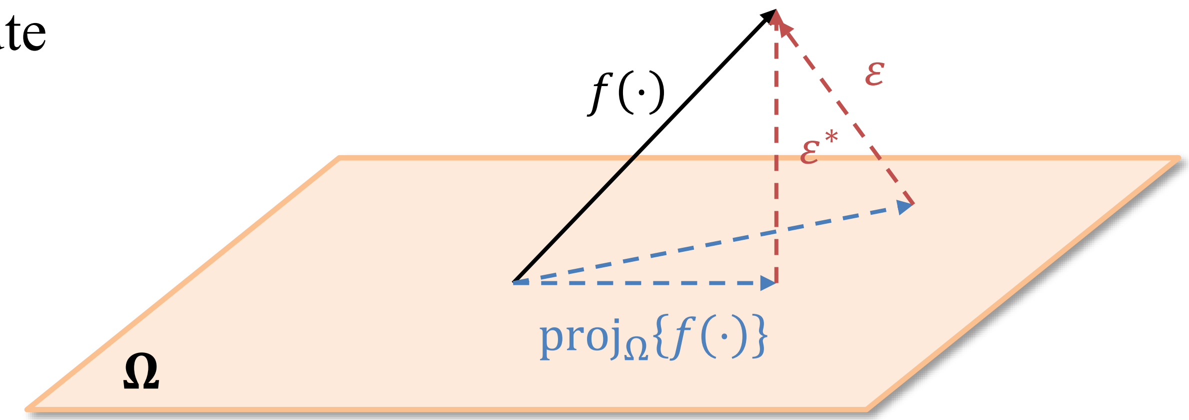
Transfer function of the end-to-end RC/ESN with W_{out}:

$$H_{RC}(z) = \sum_{k=1}^{K} W_{out}[k,k] H_k(z)$$

The Geometric Interpretation

The *non-interconnected* reservoir neurons constitute basis functions $\{H_k(z)\}$, span a subspace $\boldsymbol{\Omega}$, i.e.,

$$\boldsymbol{\Omega} = \mathrm{span}\{H_k(z)\}_{k=1}^{K}$$



Given a target function $f(\cdot)$:

- RC/ESN approximates $f(\cdot)$ as a linear combination of the basis functions
- With approximation error of ε

Considering ℓ_2 loss:

- RC/ESN finds $\{W_{out}\}$ s.t. the approximation error ε is minimized to ε^*
- RC/ESN approximates $f(\cdot)$ as the orthogonal projection onto $\boldsymbol{\Omega}$
- $\mathbf{W}_{out}$: One-shot solution of least squares

S. Jere, L. Zheng, K. Said and L. Liu, "Universal Approximation of Linear Time-Invariant (LTI) Systems Through RNNs: Power of Randomness in Reservoir Computing," in *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 2, pp. 184-198, March 2024

Wireless Channel Equalization

- A general stochastic channel system function (minimum phase) with L taps:

$$H_{ch}(z) = h_0 + h_1 z^{-1} + h_2 z^{-2} + \cdots + h_L z^{-(L-1)}$$

- Channel inverse's system function is:

$$H_{ch}^{-1}(z) = \frac{1}{\sum_{\ell=0}^{L-1} h_\ell z^{-1}} \approx \sum_{i=0}^{L-1} \frac{w_i}{1 - \alpha_i z^{-1}}$$

Poles of inverse system: α_i

- Target function: Sum of first-order IIR poles.
  - Difficulty #1: Need to approximate a sum of 1st-order IIR characterized by their respective poles α_i
  - Difficulty #2: No knowledge of locations or distribution of poles α_i.

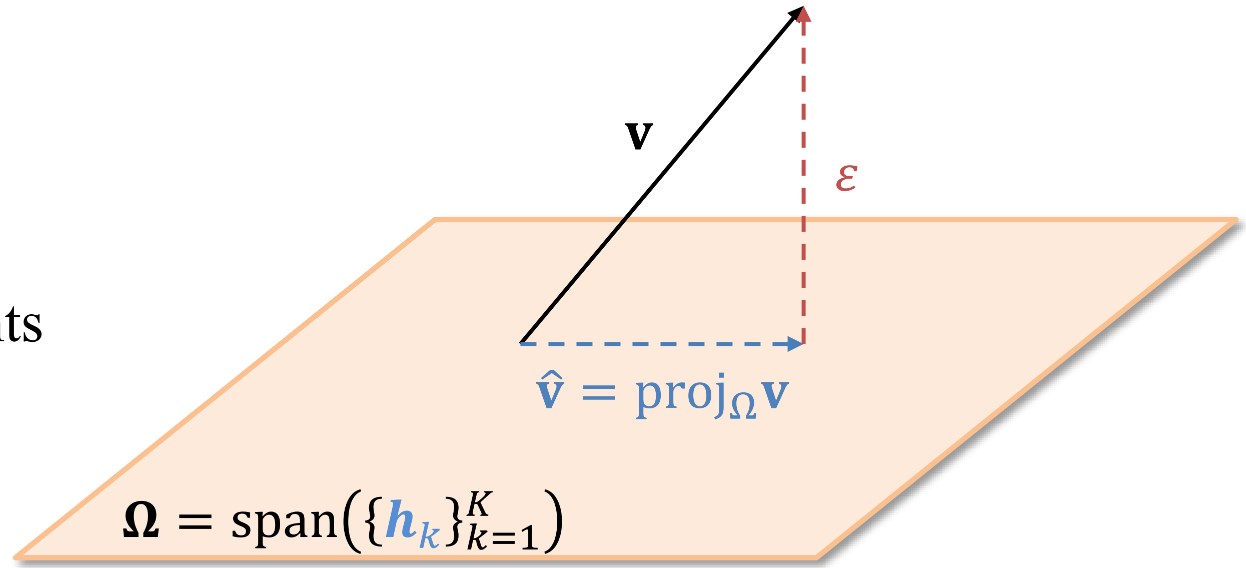
Wireless Receive Processing – Problem Formulation

Minimum phase channels:

$$H_{ch}^{-1}(e^{j\omega}) = \frac{1}{\sum_{\ell=0}^{L-1} h_\ell e^{-j\ell\omega}}$$

Sample $H_{ch}^{-1}(e^{j\omega})$ at N discrete frequency points

$$[\mathbf{v}]_n = H_{ch}^{-1}(e^{j\omega_n})$$



Each neuron generates

$$[\boldsymbol{h}_k]_n = H_k(e^{j\omega_n}) = \frac{W_{in}[k,k]}{1 - W_{res}[k,k]e^{-j\omega_n}}$$

They span a subspace $\boldsymbol{\Omega} = \mathrm{span}(\{\boldsymbol{h}_k\}_{k=1}^{K})$

RC/ESN conducts a projection of $\mathbf{v}$ onto $\boldsymbol{\Omega}$:

$$\hat{\mathbf{v}} = \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H\mathbf{v}$$

Domain Knowledge: Statistics of $\mathbf{v}$, i.e., $\mathbf{K}_{\mathbf{vv}}$

Objective: Configure $\mathbf{F}$ to minimize ε

$$\min_{\mathbf{F}} \mathbb{E}_{\mathbf{v}}[\|\mathbf{v} - \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H\mathbf{v}\|_2^2]$$

PCA Meets RC Design: Frequency-Domain Config.

- Principal Component Analysis (PCA) provides an answer
  - Perform eigen-decomposition of $\mathbf{K}_{\mathrm{vv}}$ or $\widehat{\mathbf{K}}_{\mathrm{vv}}$, i.e., $\widehat{\mathbf{K}}_{\mathrm{vv}} = \mathbf{QDQ}^H$
  - $\mathbf{Q}_M$ (first M columns of $\mathbf{Q}$), i.e., $\mathbf{Q}_M = [\boldsymbol{q}_1 \;\; \boldsymbol{q}_2 \;\; \dots \;\; \boldsymbol{q}_M]$
- Connect the matrix $\mathbf{F}$ with $\mathbf{Q}_M$: $\mathbf{F} = \mathbf{Q}_M$. F is orthogonal s.t. $\mathbf{F}^H\mathbf{F} = \mathbf{I}_M$.
  - Approximate $Q_m(e^{j\omega})$ as a strictly proper rational polynomial $R_m(e^{j\omega})$ from $\boldsymbol{q}_m$
  - Decompose $R_m(e^{j\omega})$ into first-order IIR transfer functions

$$R_m(e^{j\omega}) \approx \sum_{k=1}^{K'} \frac{q_{m,k}}{1 - p_{m,k}e^{-j\omega}}$$

- <u>Note</u>: M and K' are heuristic design choices.

Connecting with 5G (3GPP)

Statistics of $H_{ch}(z)$ or $H_{ch}(j\omega)$

Means and variances of CIR taps/clusters

Table 7.7.1-1. CDL-A

Clusters						
Cluster #	Normalized delay	Power dB	AoD °	AoA °	ZoD °	ZoA °
1	0.0000	-13.4	-178.1	51.3	50.2	125.4
2	0.3819	0	-4.2	-152.7	93.2	91.3
3	0.4025	-2.2	-4.2	-152.7	93.2	91.3
4	0.5868	-4	-4.2	-152.7	93.2	91.3
5	0.4610	-6	90.2	76.6	122	94
6	0.5375	-8.2	90.2	76.6	122	94
7	0.6708	-9.9	90.2	76.6	122	94
8	0.5750	-10.5	121.5	-1.8	150.2	47.1
9	0.7618	-7.5	-81.7	-41.9	55.2	56
10	1.5375	-15.9	158.4	94.2	26.4	30.1
11	1.8978	-6.6	-83	51.9	126.4	58.8
12	2.2242	-16.7	134.8	-115.9	171.6	26
13	2.1718	-12.4	-153	26.6	151.4	49.2
14	2.4942	-15.2	-172	76.6	157.2	143.1
15	2.5119	-10.8	-129.9	-7	47.2	117.4
16	3.0582	-11.3	-136	-23	40.4	122.7
17	4.0810	-12.7	165.4	-47.2	43.3	123.2
18	4.4579	-16.2	148.4	110.4	161.8	32.6
19	4.5695	-18.3	132.7	144.5	10.8	27.2
20	4.7966	-18.9	-118.6	155.3	16.7	15.2
21	5.0066	-16.6	-154.1	102	171.7	146
22	5.3043	-19.9	126.5	-151.8	22.7	150.7
23	9.6586	-29.7	-56.2	55.2	144.9	156.1

Per-Cluster Parameters					
Parameter	c_{ASD}	c_{ASA}	c_{ZSD}	c_{ZSA}	XPR
Unit	°	°	°	°	dB
Value	5	11	3	3	10

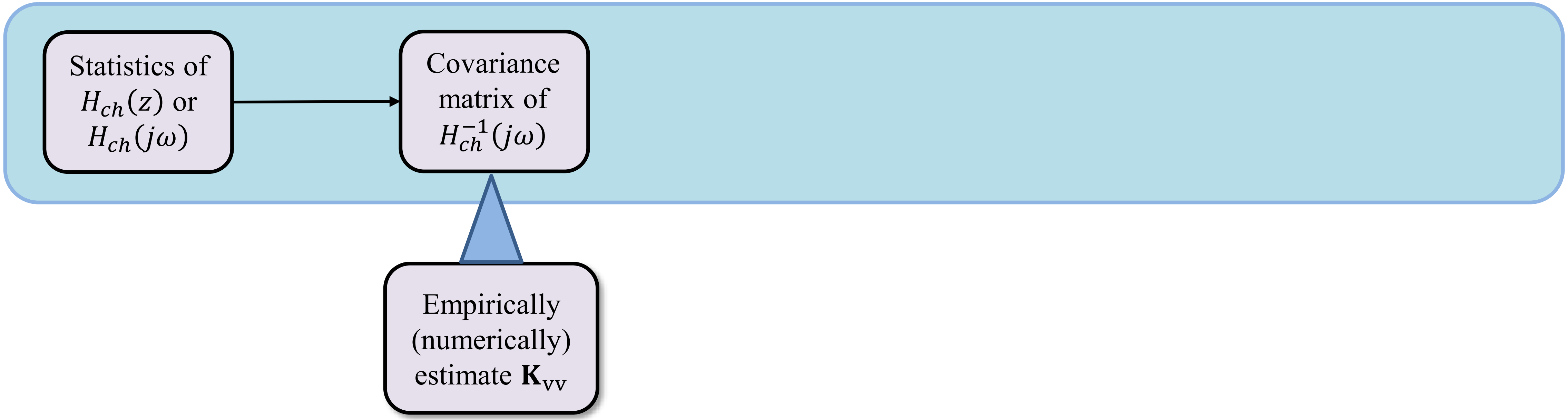
Table 7.7.2-4. TDL-D.

Tap #	Normalized delay	Power in [dB]	Fading distribution
1	0	-0.2	LOS path
	0	-13.5	Rayleigh
2	0.035	-18.8	Rayleigh
3	0.612	-21	Rayleigh
4	1.363	-22.8	Rayleigh
5	1.405	-17.9	Rayleigh
6	1.804	-20.1	Rayleigh
7	2.596	-21.9	Rayleigh
8	1.775	-22.9	Rayleigh
9	4.042	-27.8	Rayleigh
10	7.937	-23.6	Rayleigh
11	9.424	-24.8	Rayleigh
12	9.708	-30.0	Rayleigh
13	12.525	-27.7	Rayleigh

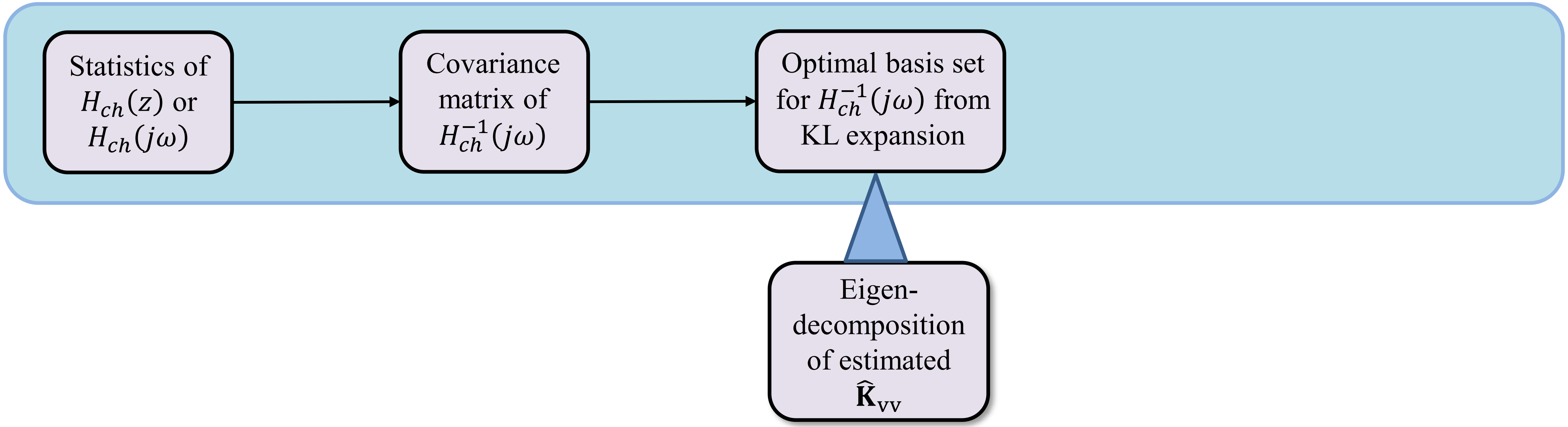
NOTE: The first tap follows a Ricean distribution with a K-factor of K_1 = 13.3 dB and a mean power of 0dB.

Ref: 3GPP Technical Report (TR) 38.901

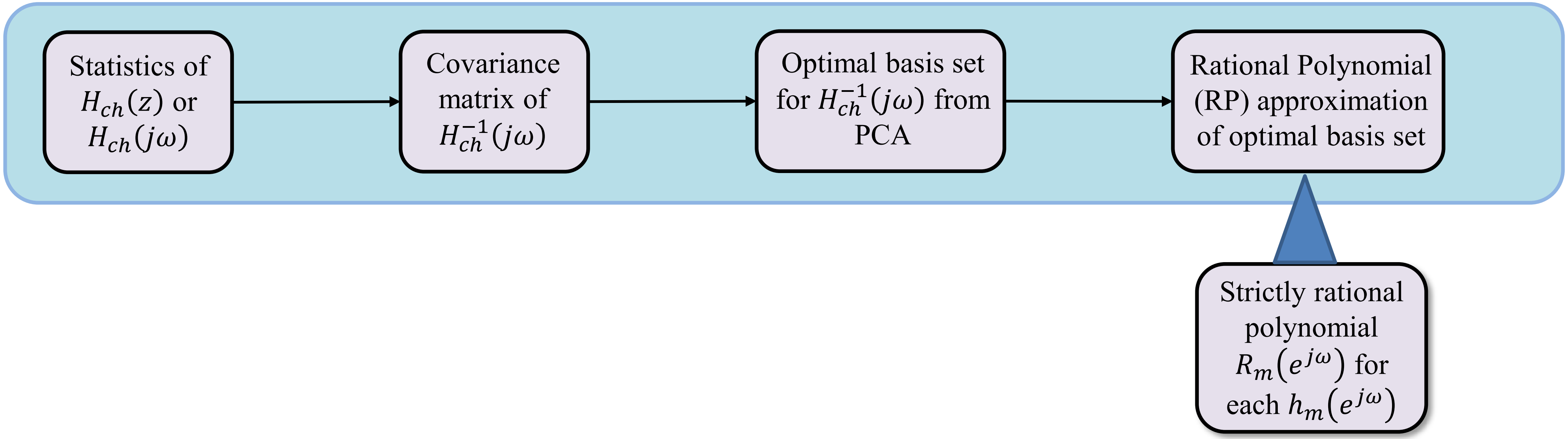
Connecting with 5G (3GPP)



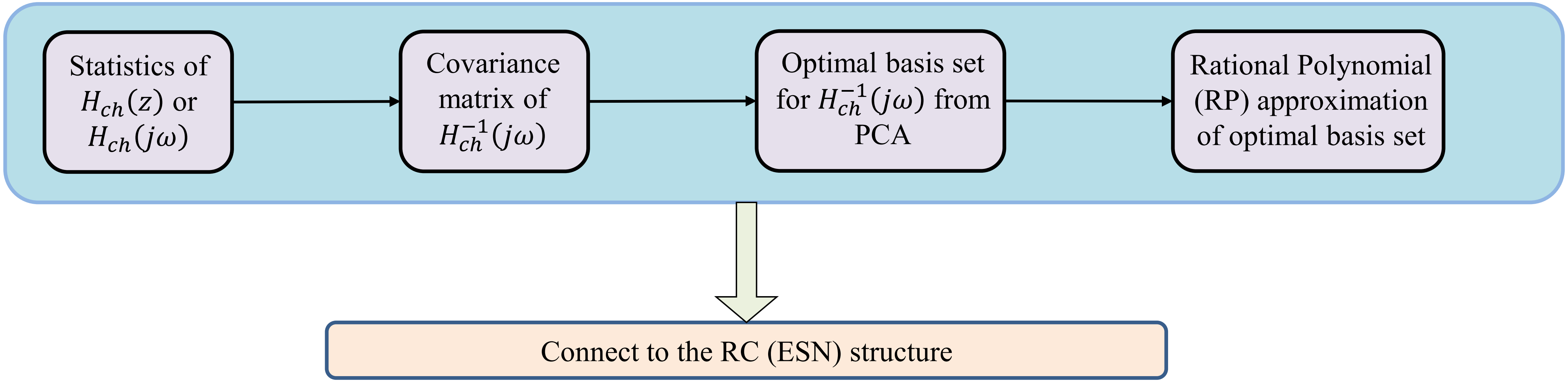
Connecting with 5G (3GPP)



Connecting with 5G (3GPP)



Connecting with 5G (3GPP)



Connecting with 5G (3GPP)

Statistics of $H_{ch}(z)$ or $H_{ch}(j\omega)$ → Covariance matrix of $H_{ch}^{-1}(j\omega)$ → Optimal basis set for $H_{ch}^{-1}(j\omega)$ from PCA → Rational Polynomial (RP) approximation of optimal basis set

Connect to the RC (ESN) structure

First-order decomposition of RP approximations:

$$R_m\left(e^{j\omega}\right) \approx \sum_{k=1}^{K_{den}} \frac{q_{m,k}}{1 - p_{m,k}e^{-j\omega}} ; m = 1, \dots , M$$

Connecting with 5G (3GPP)

Statistics of $H_{ch}(z)$ or $H_{ch}(j\omega)$ → Covariance matrix of $H_{ch}^{-1}(j\omega)$ → Optimal basis set for $H_{ch}^{-1}(j\omega)$ from PCA → Rational Polynomial (RP) approximation of optimal basis set

⇓

Configure the RC (ESN) structure

⇓

First-order decomposition of RP approximations:

$$R_m\left(e^{j\omega}\right) \approx \sum_{k=1}^{K_{den}} \frac{q_{m,k}}{1 - p_{m,k}e^{-j\omega}}; m = 1, \dots, M$$

- Configure $\mathbf{W}_{res} \in \mathbb{C}^{MK_{den} \times MK_{den}} = diag\left(\{p_{m,k}\}\right)$
- Configure $\mathbf{W}_{in} \in \mathbb{C}^{MK_{den} \times 1} = \{q_{m,k}\}$
- Total neurons in reservoir: $K = MK_{den}$

S. Jere, K. Said, L. Zheng and L. Liu, "Towards Explainable Machine Learning: The Effectiveness of Reservoir Computing in Wireless Receive Processing," *MILCOM 2023 - 2023 IEEE Military Communications Conference (MILCOM)*, Boston, MA, USA, 2023, pp. 667-672.